Module: Cloud Security 2026 HS

Dr. Andrew Hutchison

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Lecturer introduction: Andrew Hutchison

Dr. Andrew Hutchison is a cyber security specialist and has worked for companies including Google (as cloud security specialist) and previously at T-Systems Switzerland, as part of the international Telekom Security organization of Deutsche Telekom.

Andrew has extensive experience advising organisations with respect to their security strategy, roadmap and solutions. He previously held other roles in the international organization of T-Systems – in particular as Executive Advisor CyberSecurity, International Programme Executive for CyberSecurity, and as General Manager / Vice President of the Telecommunications and Security Business in South Africa.

Prior to joining T-Systems Andrew co-founded an IT consulting and training company, which was acquired by a South African IT group, and he has also been on the faculty of the Computer Science Department at the University of Cape Town (UCT) as an Adjunct Professor. In 2020 he was invited to establish a course in cloud security for Hochschule Luzern.

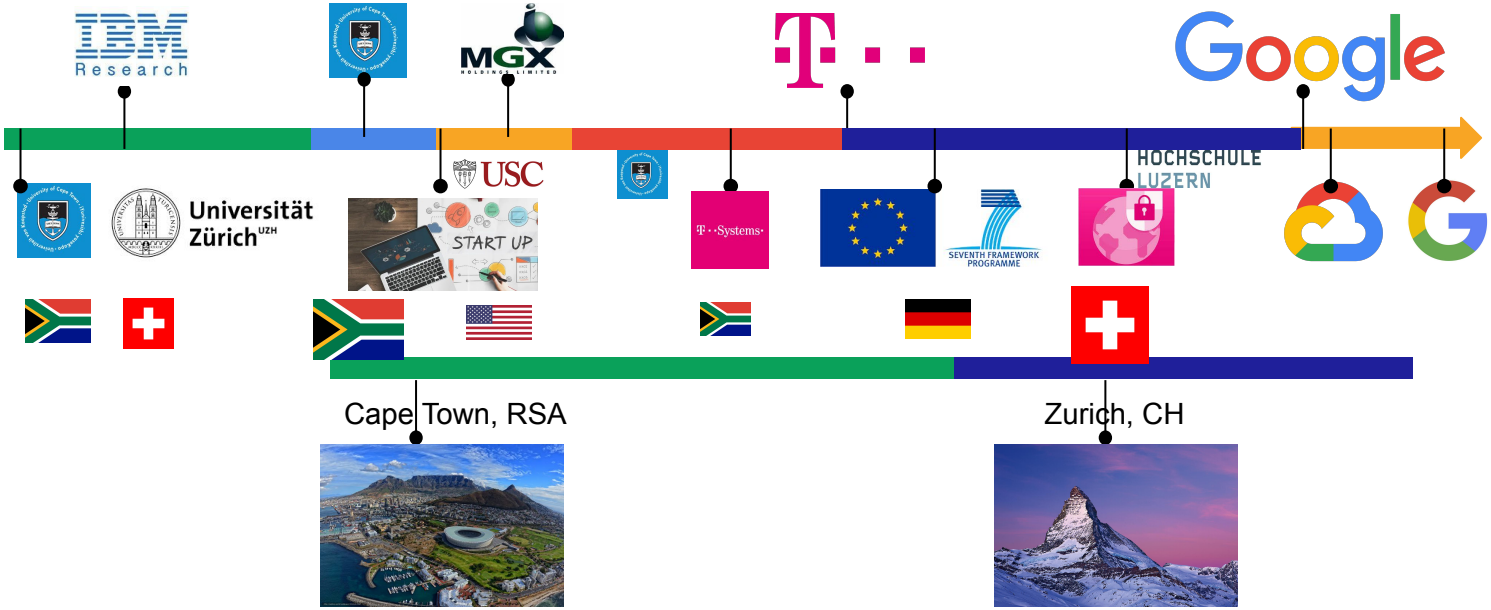
Andrew has a PhD in Computer Science (Information Security), which he completed at the University of Zürich, Switzerland while working at the IBM Zürich Research Laboratory in Rüschlikon. He is also a Google Cloud Certified Professional Cloud Security Engineer & Cloud Digital Leader.





My Professional Experience Line and Locations …





Google Cloud

Cloud Security Module Overview

The module **Cloud Security** covers aspects of Cloud Technology + Operation & Cloud Management with respect to Security.
Starting with a consideration of security layers and responsibilities (around IaaS / PaaS and SaaS type services) the technology topics continue to consider identity in the cloud and how to align this with existing enterprise directory services. Encryption in the cloud is considered, for data in transit, at rest and while being processed. The security of microservice unit containers is reviewed including how to direct containers to relevant execution environments according to security posture and guidelines. Network and Data Traffic control looks at network and traffic security issues connecting to a cloud service, and within the cloud environment between different elements. Monitoring, vulnerability assessment and overall visibility are important factors especially in the context of overall enterprise management and this forms a further topic for the technology and operations aspect. Finally security blueprints & reference architectures are presented and evaluated to guide secure cloud implementations. Issues around security of AI models and securely using ML/AI on cloud will also be introduced.

From a cloud management perspective, the topic of cloud strategy will be addressed together with processes / roles and responsibilities to ensure a secure operation. The risk management environment and associated control frameworks will be introduced. Compliance management will introduce the most important industry standard certifications and assessments which cloud providers should have. Supplier management will cover assessment of cloud providers and the important topics of privacy and data protection including Swiss cloud usage guidelines, GDPR and data residency issues.

Cloud Security Module Assessment

- The CloudSec module will be assessed via:
  - a Multiple Choice Question (MCQ) exam
  - combined sections (technical + management)
  - 90 mins
  - “open book” but without Internet access

*** Further details on the exam format and scheduling will be communicated by Sami Hassanein, course lead, and any questions in this regard should be directed to him please!*

Cloud Security Module Resources

- A folder will be created on Illias for each lecture topic
- Resources for each topic will typically include:
  - The slides of the material
  - Recorded presentation of the lecture material
  - Readings and/or other materials (links) relevant to the topic

Cloud Security Module Structure

Cloud Technology and Operation

1. Security Layers and Responsibilities
2. Identity in Cloud
3. Encryption in the Cloud
4. Container Security
5. Network and Data Traffic Control
6. Security Visibility and Monitoring
7. Security Blueprints

Cloud Management

1. Cloud Strategy
2. Processes, Roles, Responsibilities
3. Risk Management
4. Control Framework
5. Compliance Management
6. Supplier Management
7. Privacy and Data Protection

Cloudsec Module Overview

DATE	TRACK	LECTURER
Fri 18th Sept	Technical 1	Andrew *Rotkreuz + Online*
Fri 25th Sept	Technical 2	Andrew
Fri 2nd Oct	Technical 3	Andrew
Fri 9th Oct	Technical 4	Andrew
Fri 16th Oct	Technical 5	Andrew / *Recording*
Fri 23rd Oct	Technical 6	Andrew
Fri 30th Oct	Technical 7	Andrew
Fri 6th Nov	Management 1	Sami
Fri 13th Nov	Management 2	Sami
Fri 20th Nov	Management 3	Sami
Fri 27th Nov	Management 4	Sami
Fri 4th Dec	Management 5	Sami
Fri 11th Dec	Management 6	Sami
Fri 18th Dec	Management 7	Sami

LECTURE TOPIC / SCHEDULING 2024	SUB-TOPICS
Lecture 1: **Security Layers and Responsibilities**	· IaaS / PaaS / SaaS · Hybrid and MultiCloud · Landing zone concept and models of AWS, Azure, GCP · Shared responsibility model
Lecture 2: **Identity in Cloud and Mapping back to Organisational Directories**	· Cloud IAM · OAuth2.0 / SAML
Lecture 3: **Encryption in the Cloud**	· Data at Rest / in Transit / in Processing · Customer Managed Encryption Keys (BYOK) / Hardware Encryption Modules e.g. Thales · Tokenization and Anonymization / DLP · Crypto Key Justifications
Lecture 4: **Container Security**	· Verified binaries (binary authorization) · CI/CD integration of security · Vulnerability scanning
Lecture 5: (recording) **Network and Data Traffic Control**	· Cloud Firewalls · Cloud Identity Aware Proxy · IP address range considerations
Lecture 6: **Security Visibility and Monitoring**	· Cloud Console · Security Information and Event Monitoring (SIEM) integration e.g. Splunk, HP Arcsight, IBM QRadar, Google SecOps etc.
Lecture 7: **Security blueprints and reference architectures**	· Secure AI Frameworks · Best practice Frameworks and security epics

Readings: What is cloud computing
- AWS Overview
- AWS Security Learning
- AWS Overview Security Processes
- Google Infrastructure Whitepaper
- Google Security Whitepaper
- Landing zone design in GCP
- Landing zone in Azure
- Landing zone in AWS
- Security Megatrends

Lecture 1: Security Layers and Responsibilities

So You Want to Be a Cloud Data Security Engineer?!

- Are you passionate about protecting data in the cloud and have strong IT Security knowledge?
  Do you enjoy keeping your fingers on the pulse of technology?

  Within our IT Security division in Zurich we are looking for a Cloud Data Security Engineer who enjoys working in all domains of Data Security, sharing new ideas and is keen on driving continuous improvement at all times.Your opportunity

  Engineering and building up Data Security solutions in the area of:
  - Big Data Security
  - Data Leakage Prevention (DLP)
  - Information Rights Management (IRM)
  - Cloud Data Security
  - General Data Protection Regulation topics (GDPR)
  - Managing IT Security infrastructure projects in close coordination with stakeholders from the various business sectors
  - Consulting and supporting application managers as well as internal or external project managers on Data Security topics
  - Application responsibility for IT Security infrastructure components including operation, monitoring and 3rd level support as well as on-call service

Swiss Job Advert, LinkedIn

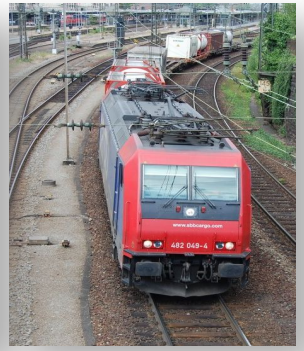
Digitization and AI Transforming Industries







Smart logistics



Predictive Analytics



Smart Factory / IOT



Smart Data



Rail & Public Transport



Digital Store

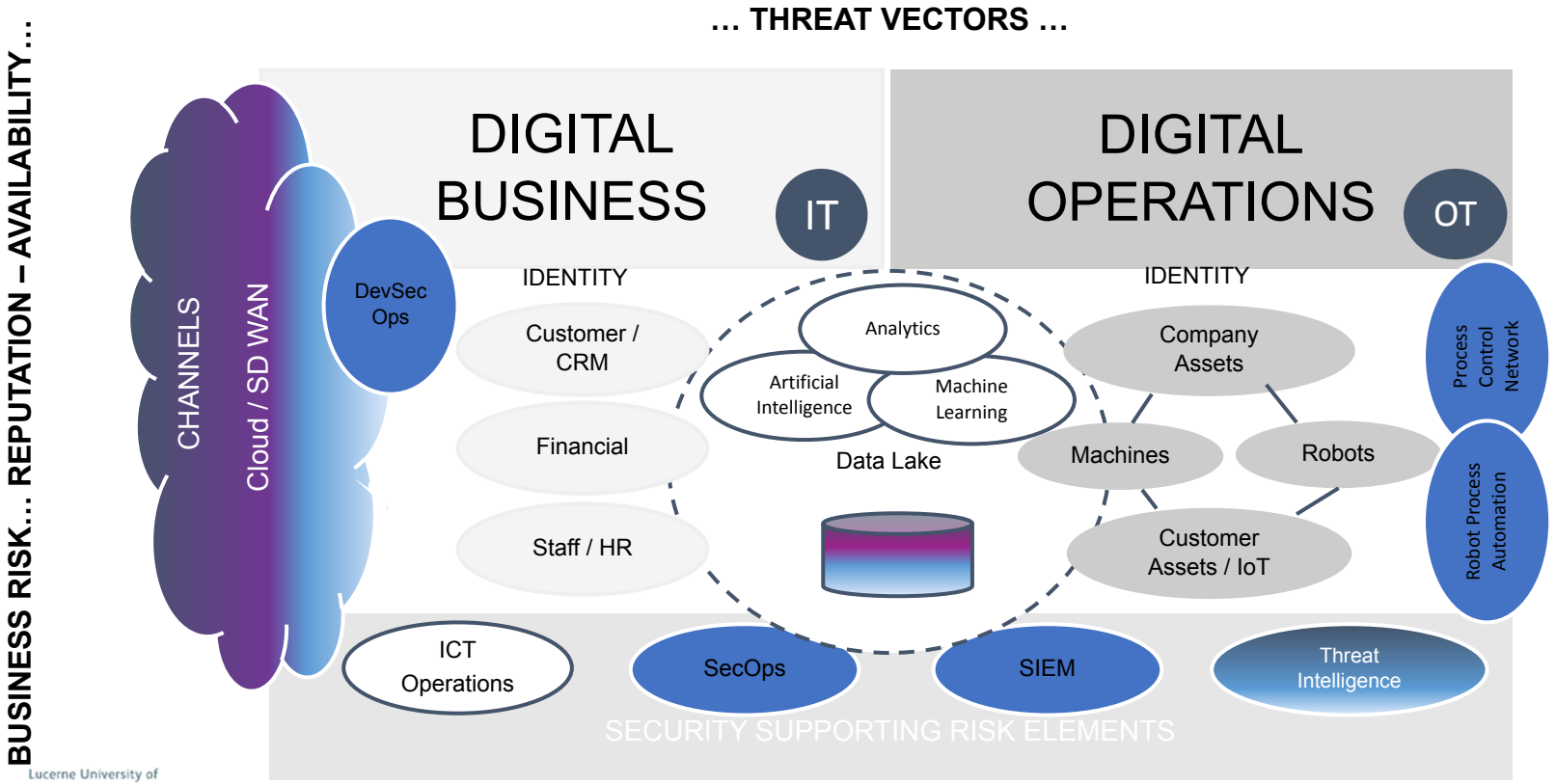


FINANCIAL SERVICES

Applied Sciences and Arts

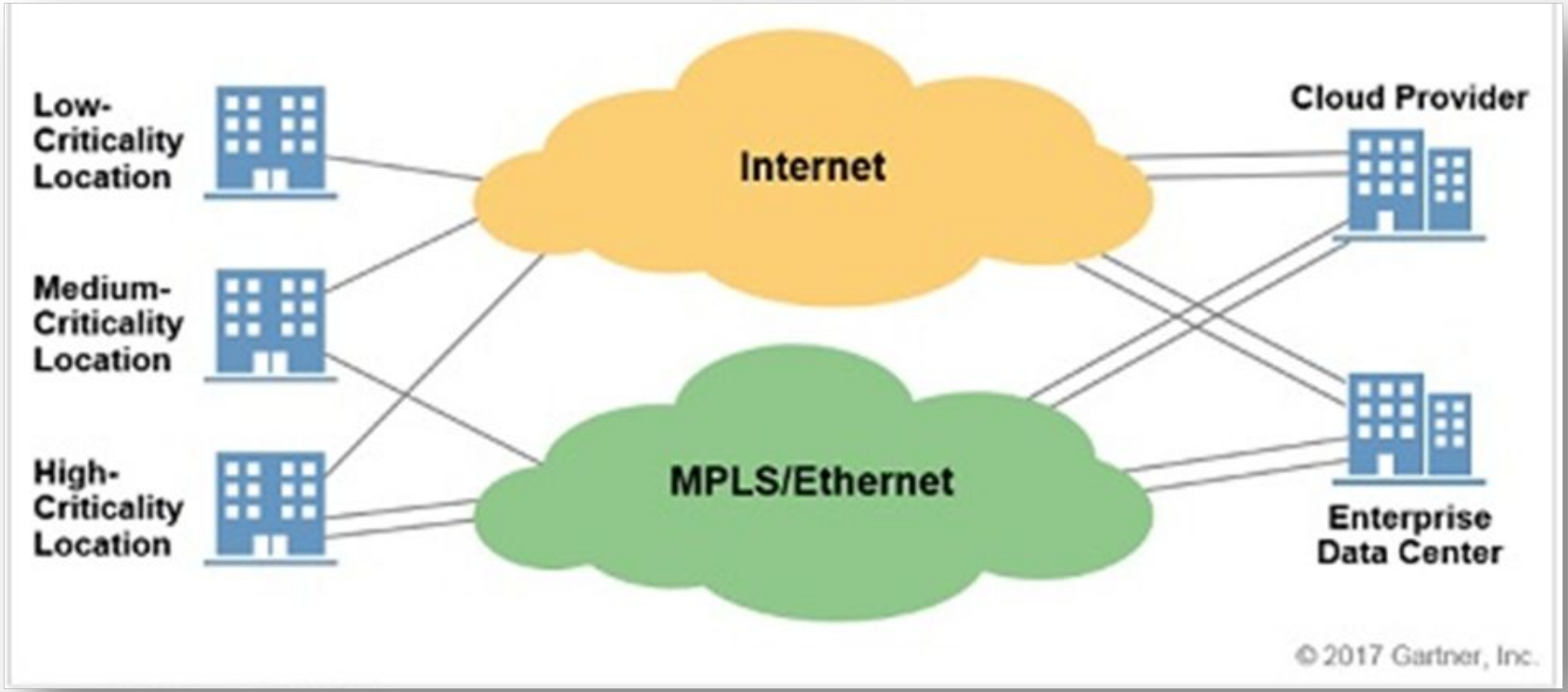
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Security Context View of Digital Transformation



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Evolving World of Network & Processing



What is Cloud Computing?

Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources through a cloud services platform via the Internet with pay-as-you-go pricing. Whether you are running applications that share photos to millions of mobile users or you're supporting the critical operations of your business, a cloud services platform provides rapid access to flexible and low-cost IT resources. With cloud computing, you don't need to make large upfront investments in hardware and spend a lot of time on the heavy lifting of managing that hardware. Instead, you can provision exactly the right type and size of computing resources you need to power your newest bright idea or operate your IT department. You can access as many resources as you need, almost instantly, and only pay for what you use.

Cloud computing provides a simple way to access servers, storage, databases and a broad set of application services over the Internet. A cloud services platform such as Amazon Web Services owns and maintains the network-connected hardware required for these application services, while you provision and use what you need via a web application.

Source: Overview of Amazon Web Services, AWS WhitePaper

Information Lifecycle



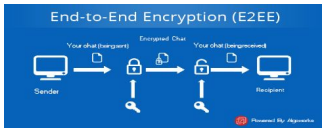
Source: sibenco.com

What Are Our Security Goals For Cloud Services? Back to First Principles …

	Security Service	Domain
1	**Authentication** (who created or sent the data)	
2	**Access Control** (prevent misuse of resources)	
3	**Confidentiality** (privacy)	
4	**Integrity** (has not been altered)	
5	**Non-Repudiation** (undeniable send/receive)	
+	**Availability** (access to systems)	

Cloud





Cloud Computing Models

Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) contains the basic building blocks for cloud IT and typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space. IaaS provides you with the highest level of flexibility and management control over your IT resources and is most similar to existing IT resources that many IT departments and developers are familiar with today.

Platform as a Service (PaaS)

Platform as a Service (PaaS) removes the need for your organization to manage the underlying infrastructure (usually hardware and operating systems) and allows you to focus on the deployment and management of your applications. This helps you be more efficient as you don't need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.

Software as a Service (SaaS)

Software as a Service (SaaS) provides you with a completed product that is run and managed by the service provider. In most cases, people referring to Software as a Service are referring to end-user applications. With a SaaS offering you do not have to think about how the service is maintained or how the underlying infrastructure is managed; you only need to think about how you will use that particular piece of software. A common example of a SaaS application is web-based email which you can use to send and receive email without having to manage feature additions to the email product or maintain the servers and operating systems that the email program is running on.

Source: Overview of Amazon Web Services, AWS WhitePaper

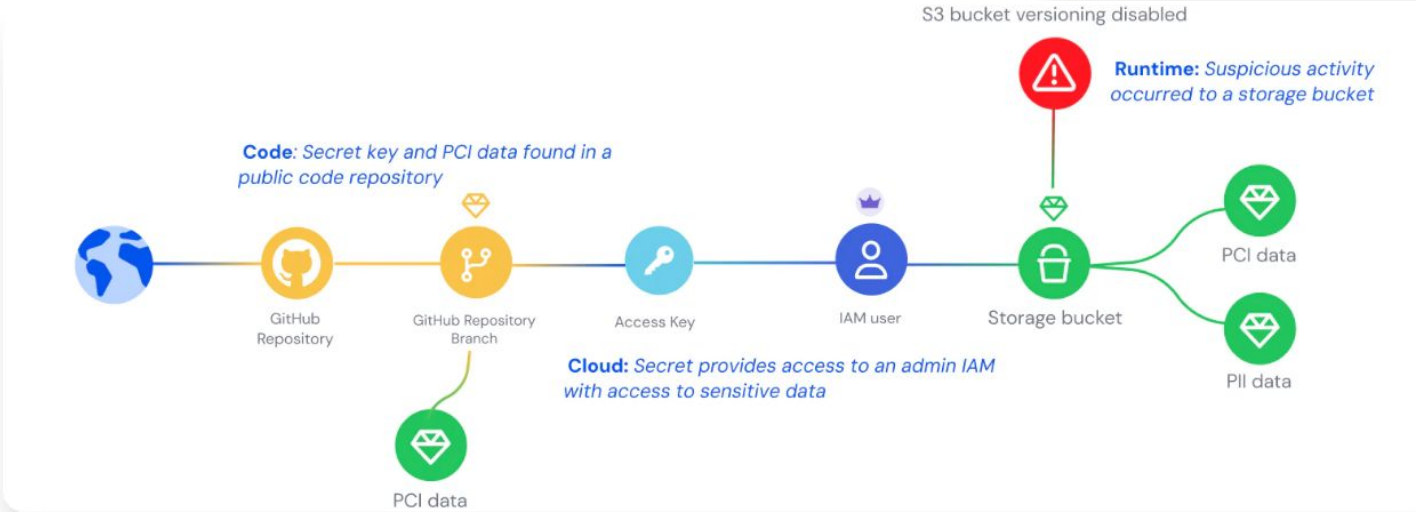
Authority Model for Various Models of Cloud Offering

	IaaS	PaaS	SaaS
	Overall system co-ordinated by cloud provider	Cloud provider manages VMs including OS and platform software (e.g. database software etc). Tenant isolation.	Control over the overall system shifts even further to Cloud Provider
Top Level Admin	Cloud Provider	Cloud Provider	Cloud Provider
Second Level Admin	Customer	Cloud Provider	Cloud Provider
Third Level Admin	• Cloud Provider Employee designated as support manager (access granted by Customer)	Customer	Cloud Provider
Eligible Admin		Cloud Provider Employee designated to provide tenant-related, dedicated support (access granted by Customer)	

Source: SBA: Cloud Guidelines: A guide to secure cloud banking (PDF

(Toxic) Combinations of Risk

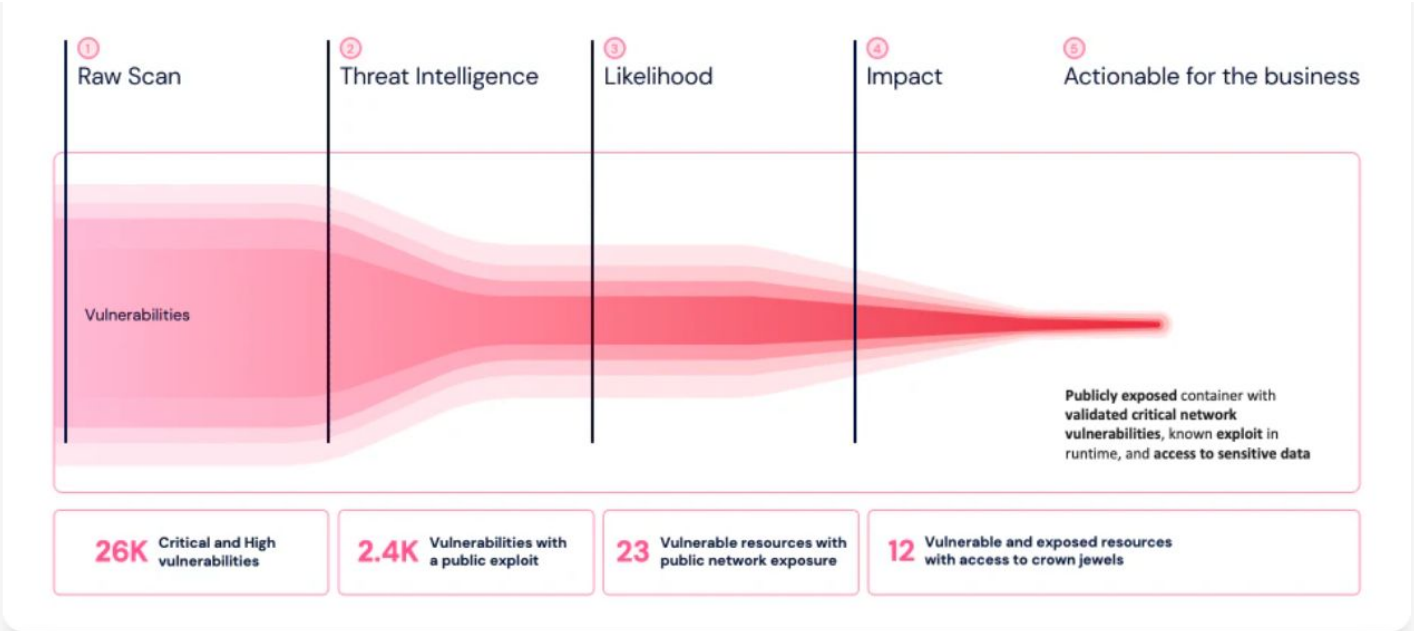
Severity of Issues		
	Critical	A clear and present danger to your organization's data and/or cloud infrastructure
	High	A serious danger, but the potential damage is smaler.
	Medium	A theoretical danger, but malicious actors would have difficulty exploiting the opening, or the potential damage is minor.
	Low	A best practice more than an actual danger.
	Info	A theoretical danger, but malicious actors would have difficulty exploiting the opening, or the potential damage is minor.



The Wiz Security Graph visualizes your cloud stack to identify the risks in each layer and deliver actionable insights.

https://www.wiz.io/blog/the-anatomy-of-a-toxic-combination-of-risk

Risk-based approach to vulnerability management in the cloud



https://www.wiz.io/blog/the-anatomy-of-a-toxic-combination-of-risk

Quiz 1

- Think of / find an example of an IaaS service
- Apply the security goals introduced to the security of a mobile phone .. how would these be relevant / realized?
- Use chatGPT, gemini, claude (or your favorite LLM!) to find examples of *toxic combinations* in cloud security



Cloud Deployment Models

Cloud

A cloud-based application is fully deployed in the cloud and all parts of the application run in the cloud. Applications in the cloud have either been created in the cloud or have been migrated from an existing infrastructure to take advantage of the benefits of cloud computing. Cloud-based applications can be built on low-level infrastructure pieces or can use higher level services that provide abstraction from the management, architecting, and scaling requirements of core infrastructure.

Hybrid

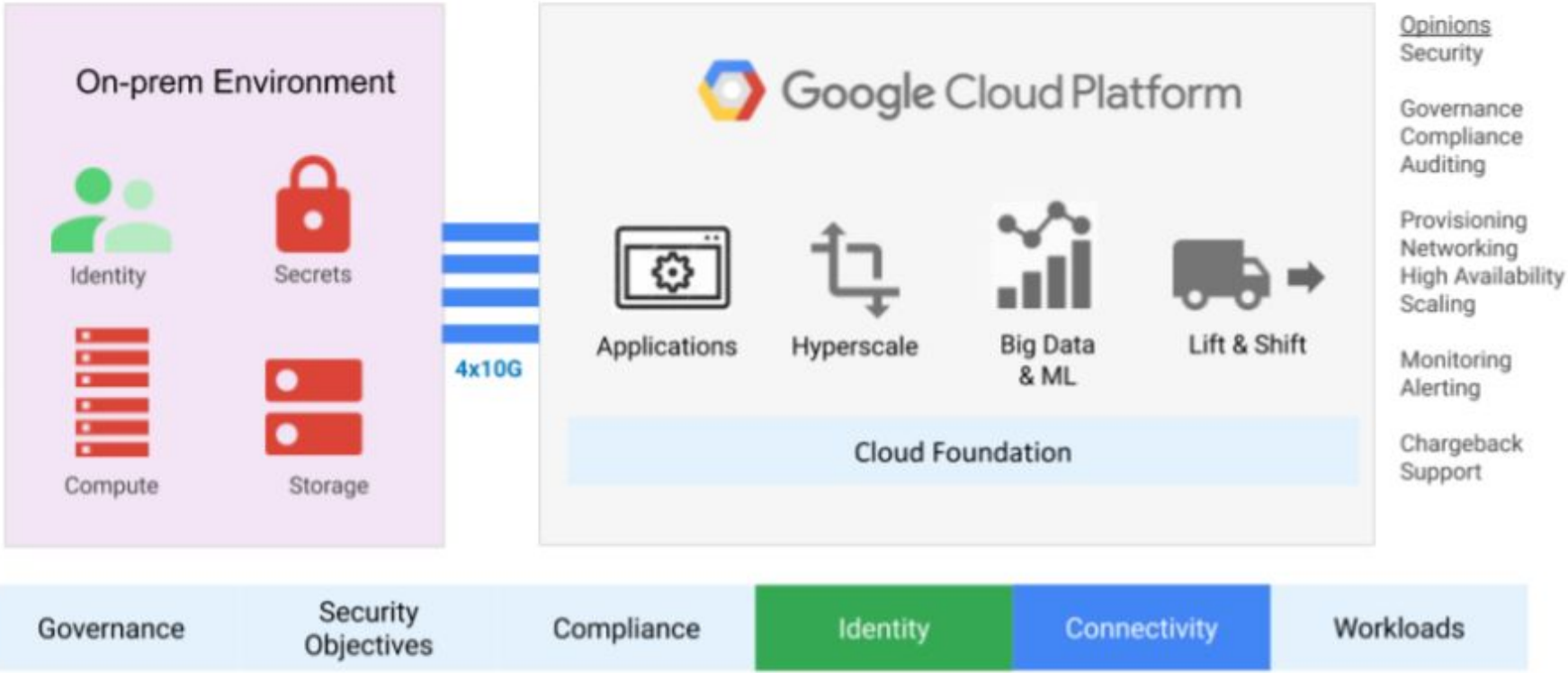
A hybrid deployment is a way to connect infrastructure and applications between cloud-based resources and existing resources that are not located in the cloud. The most common method of hybrid deployment is between the cloud and existing on-premises infrastructure to extend, and grow, an organization's infrastructure into the cloud while connecting cloud resources to the internal system. For more information on how AWS can help you with your hybrid deployment, visit our Hybrid Cloud with AWS page.

On-premises

The deployment of resources on-premises, using virtualization and resource management tools, is sometimes called the "private cloud." On-premises deployment doesn't provide many of the benefits of cloud computing but is sometimes sought for its ability to provide dedicated resources. In most cases this deployment model is the same as legacy IT infrastructure while using application management and virtualization technologies to try and increase resource utilization. For more information on how AWS can help, see Use case: Cloud services on-premises.

Source: Overview of Amazon Web Services, AWS WhitePaper

Cloud Hybrid Environment



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Source: Google, Google-Cloud-Security-Foundations-Guide.pdf

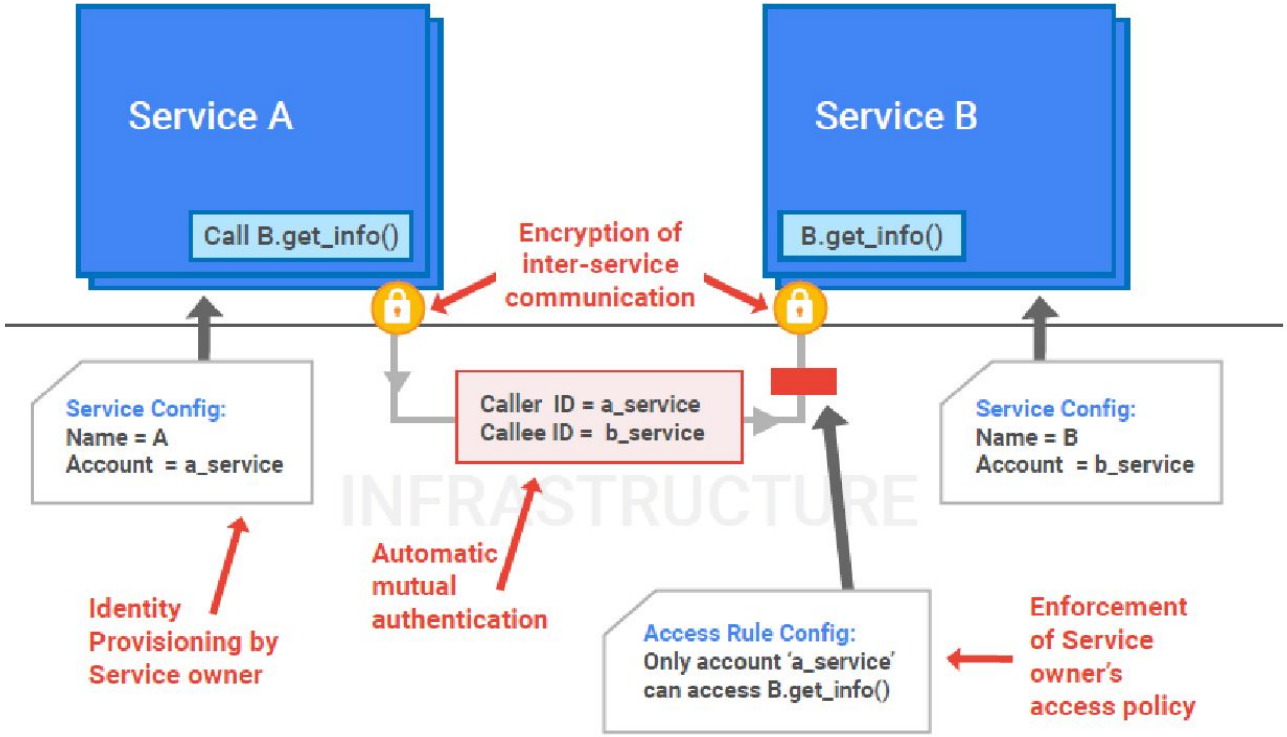
Cloud Infrastructure Layers



Source: Google, Google_infrastructure_whitepaper_fa.pdf

Example Secure Service Deployment on Cloud

Service Identity and Access Management: The infrastructure provides service identity, automatic mutual authentication, encrypted inter-service communication and enforcement of access policies defined by the service owner.

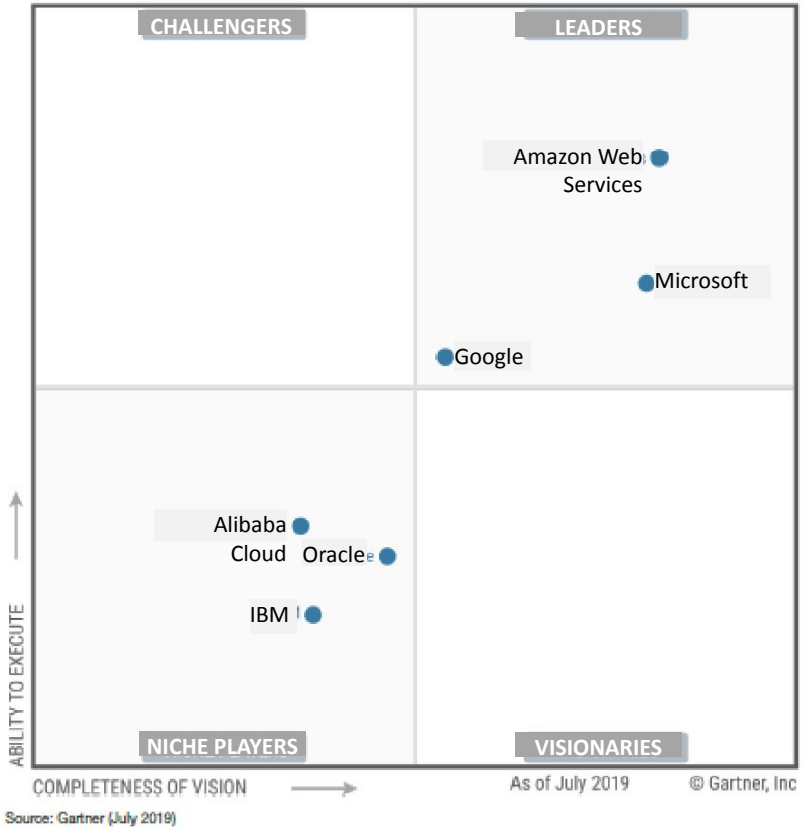


Source: Google, Google_infrastructure_whitepaper_fa.pdf

MQ for Cloud Infrastructure as a Service, Worldwide

Magic Quadrant (MQ), 2019

Cloud Service Provider (CSP)



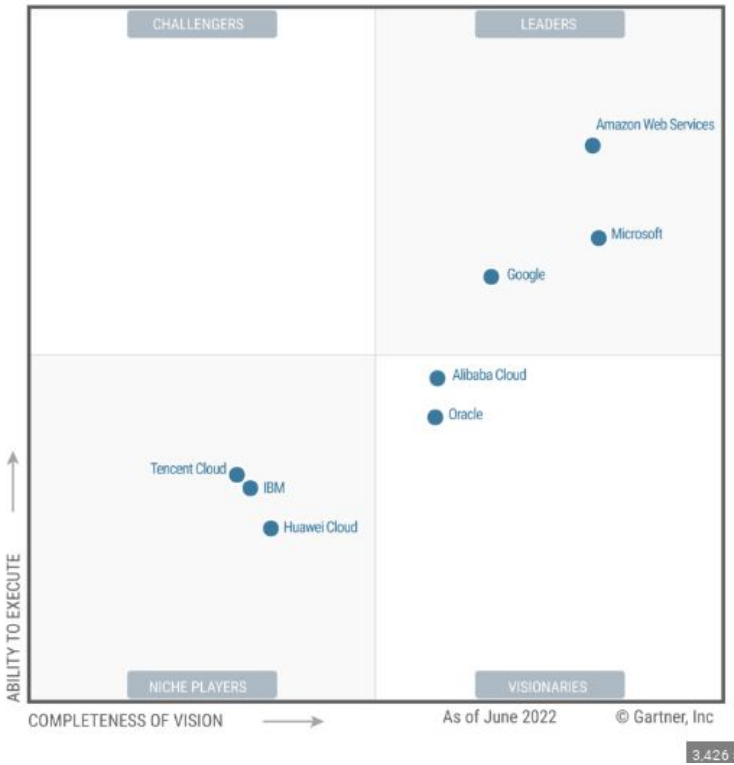
Source: Gartner

MQ for Cloud Infrastructure as a Service, Worldwide

Magic Quadrant (MQ), 2022

Cloud Service Provider (CSP)

Figure 1: Magic Quadrant for Cloud Infrastructure and Platform Services



Source: Gartner

MQ for Strategic Cloud Platform Services

Magic Quadrant (MQ), 2023

Figure 1: Magic Quadrant for Strategic Cloud Platform Services



https://cloud.google.com/blog/products/infrastructure-modernization/google-is-a-leader-in-2023-gartner-mq-for-scps

Source: Gartner

MQ for Strategic Cloud Platform Services

Magic Quadrant (MQ), 2025

Figure 1: Magic Quadrant for Strategic Cloud Platform Services



Source: Gartner

MQ for Strategic Cloud Platform Services

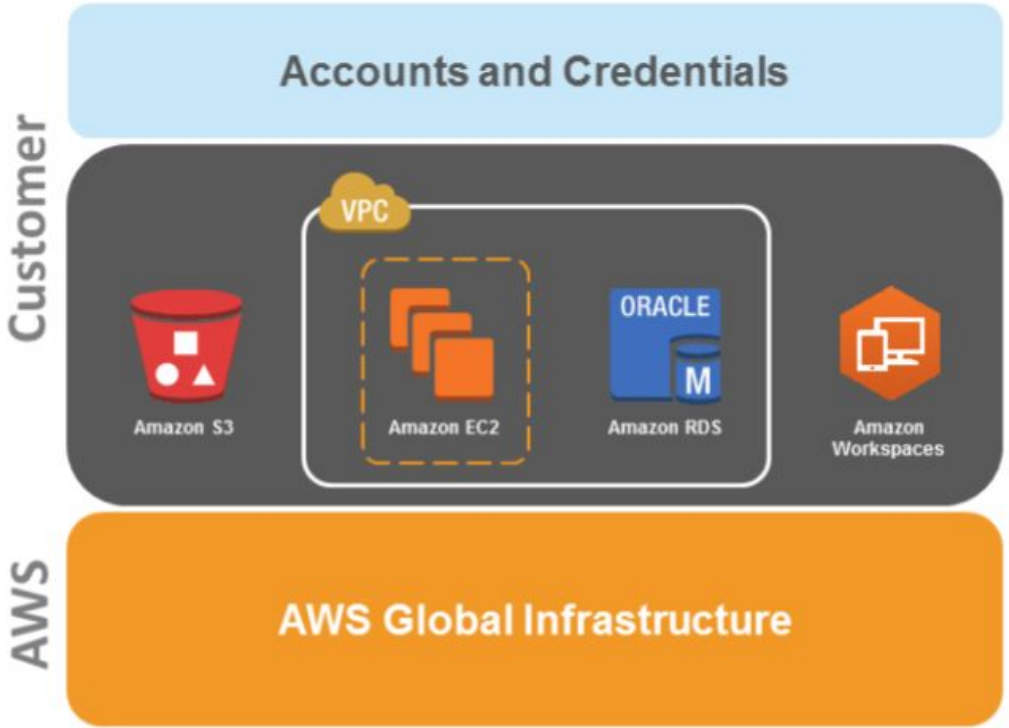
Magic Quadrant (MQ), 2026

Figure 1: Magic Quadrant for Strategic Cloud Platform Services



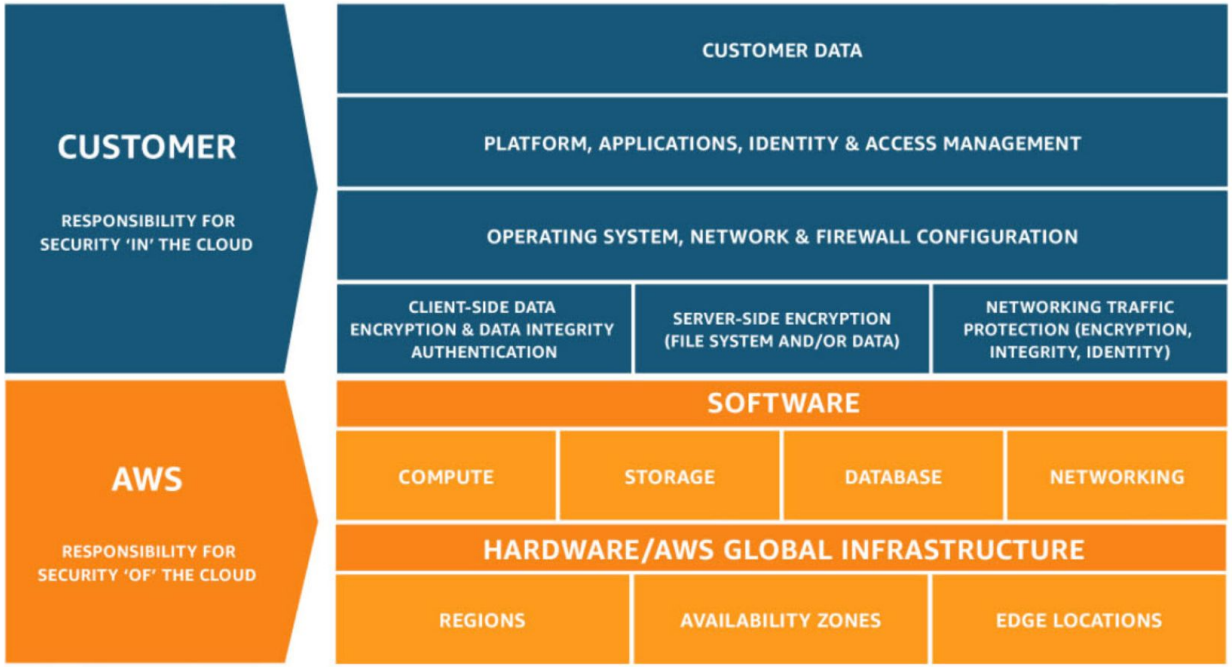
Source: Gartner

AWS Shared Responsibility Model - Overview



Source: Amazon

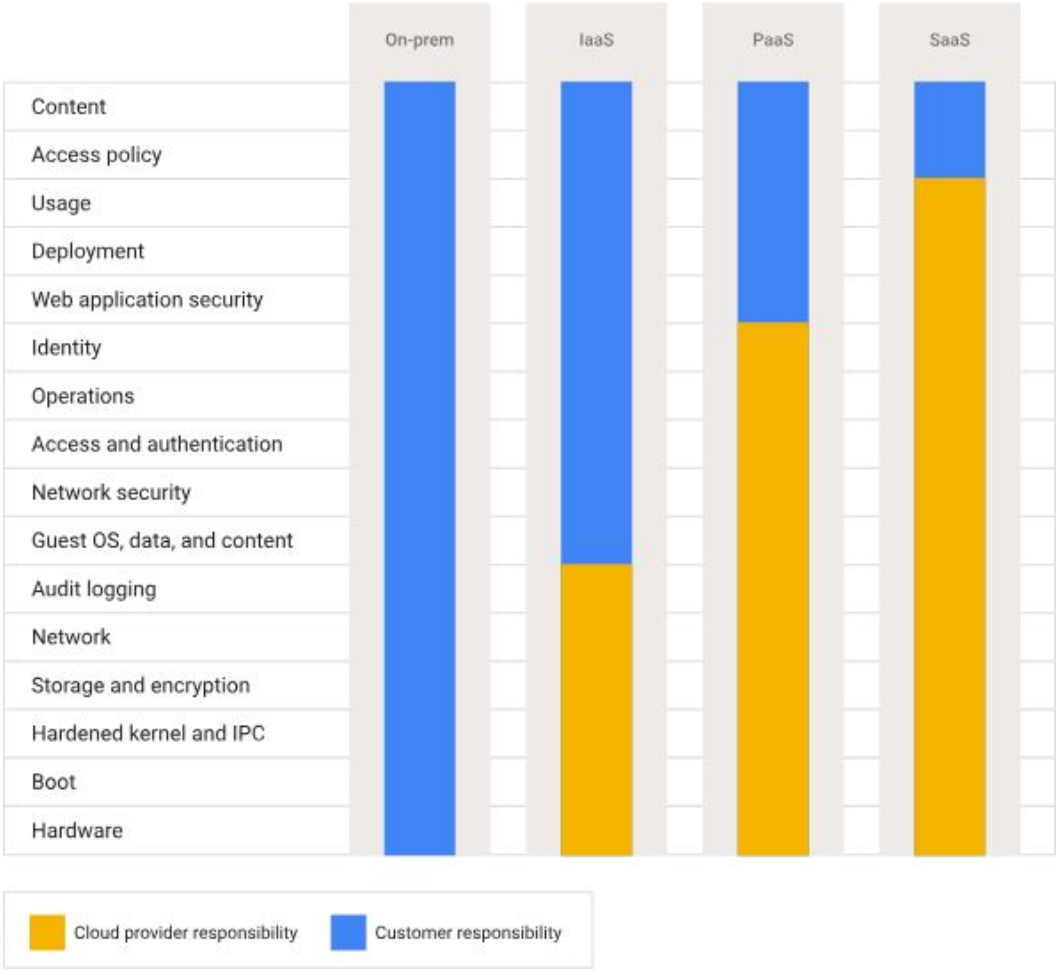
AWS Shared Responsibility Model - Detailed



Customer implements security to protect own content, platform, applications, systems and networks

Source: https://aws.amazon.com/compliance/shared-responsibility-model/

Google Shared Responsibility Model



Source: https://docs.cloud.google.com/architecture/framework/security/shared-responsibility-shared-fate?hl=en

Quiz 2

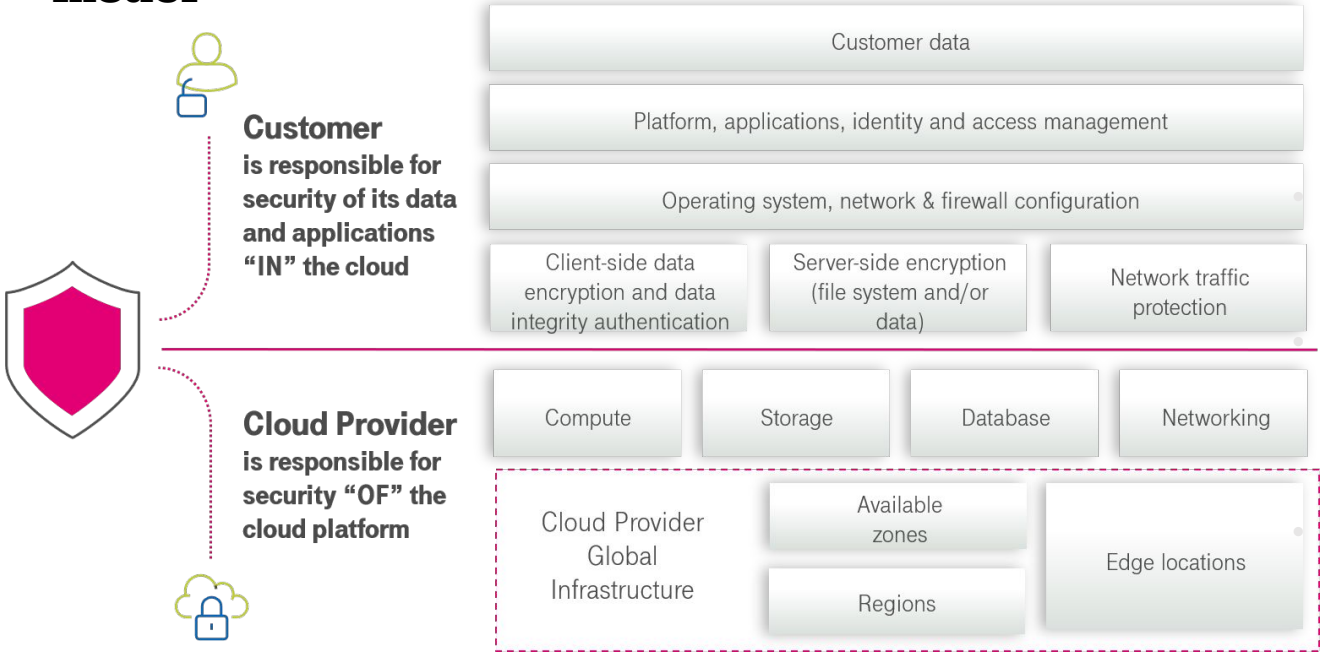
- What is the difference between hybrid cloud and multi-cloud?
- What is meant by the shared responsibility model?
- Is it possible for the user of a SaaS cloud model to get event logging information?
- Use your favourite LLM to build a table Comparing the AI services offered on GCP, AWS and Microsoft Azure



Cloud Security Implications - Analyst

> **Through 2022, 95% of cloud security failures will be the customer's fault.**
>
> Gartner

Shared responsibility model



A proper architecture, strategy & processes are required

Technology & processes must be adapted to address the dynamics of the cloud

Based on specific requirements and IT landscape a tailored approach is needed

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Cloud services: GCP as example

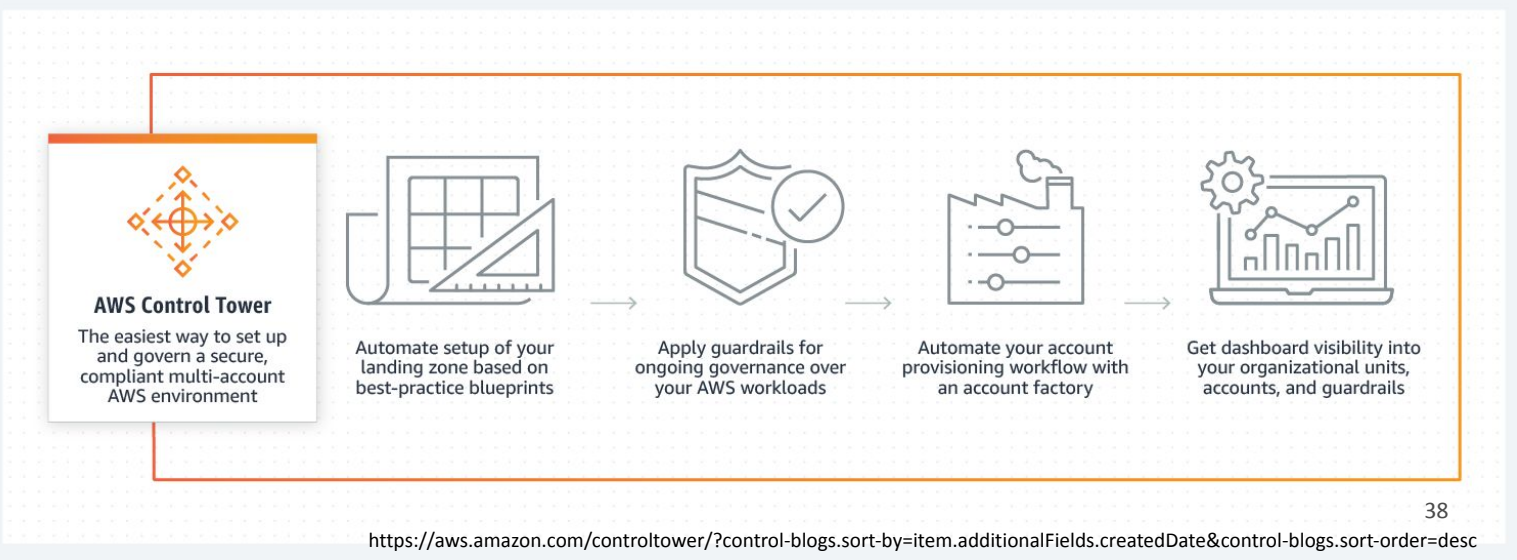
GCP Services Overview:
https://cloud.google.com/products



Landing Zone in AWS

AWS services overview:
https://aws.amazon.com/hpc/solution-components/

If you have multiple AWS accounts and teams, cloud setup and governance can be complex and time consuming, slowing down the very innovation you're trying to speed up.
AWS Control Tower provides the easiest way to set up and govern a secure, multi-account AWS environment, called a landing zone. It creates your landing zone using AWS Organizations, bringing ongoing account management and governance as well as implementation best practices based on AWS's experience working with thousands of customers as they move to the cloud. Builders can provision new AWS accounts in a few clicks, while you have peace of mind knowing that your accounts conform to company policies. Extend governance into new or existing accounts, and gain visibility into their compliance status quickly. If you are building a new AWS environment, starting out on your journey to AWS, or starting a new cloud initiative, AWS Control Tower will help you get started quickly with built-in governance and best practices.



https://aws.amazon.com/controltower/?control-blogs.sort-by=item.additionalFields.createdDate&control-blogs.sort-order=desc

Landing Zone in Azure

Azure services overview:
https://azure.microsoft.com/en-us/products/

https://www.microsoft.com/en-us/videoplayer/embed/RE4xdvm?postJsllMsg=true

- … the output of a multi-subscription Azure environment that accounts for scale, security governance, networking, and identity.
- … enables application migration, modernization, and innovation at enterprise-scale in Azure.
- … considers all platform resources that are required to support the customer's application portfolio and doesn't differentiate between infrastructure as a service or platform as a service.

"A landing zone is an environment for hosting your workloads, pre-provisioned through code."

Tailor the Azure landing zone architecture to meet requirements for further information.

See: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

Landing Zone in Azure

Scalable and modular

No single solution fits all technical environments. However, a few Azure landing zone implementation options can help you meet the deployment and operations needs of your growing cloud portfolio.

- Scalable: All Azure landing zones support cloud adoption at scale by providing repeatable environments, with consistent configuration and controls, regardless of the workloads or Azure resources deployed to each landing zone instance.
- Modular: All Azure landing zones provide an extensible approach to building out your environment, based on a common set of design areas. The extensibility of an Azure landing zone enables an organization to easily scale specific elements of the environment, as requirements evolve.

See: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

Landing Zone in Azure

Platform vs. application landing zones

- Platform landing zones: Subscriptions deployed to provide centralized services, often operated by a central team, or a number of central teams split by function (e.g. networking, identity), which will be used by various workloads and applications. Platform landing zones represent key services that often benefit from being consolidated for efficiency and ease of operations. Examples include networking, identity, and management services.
- Application landing zones: One or more subscriptions deployed as an environment for an application or workload. Application landing zones are placed in management groups like 'corp' or 'online' beneath the 'landing zones' management group to ensure policy controls are correctly applied. Application landing zones can be subcategorized as follows:
    - Centrally managed: A central IT team fully operates the landing zone. The team applies controls and platform tools to both the platform and application landing zones.
    - Technology platforms: With technology platforms such as AKS or AVS, the underlying service is often centrally managed. The applications running on top of the service have delegated responsibilities to application teams. This results in modified controls or access permissions compared to centrally managed landing zones.
    - Workload: A platform administration team delegates the entire landing zone to a workload team to fully manage and support the environment; whilst still being controlled by the policies applied from the Management Groups above that the platform team control. This might include adding additional policies at the subscription scope and using alternative tooling for deploying, securing or monitoring workloads that is fully controlled and operated by the workload team.

See: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

Landing Zone in Azure

https://www.microsoft.com/en-us/videoplayer/embed/RE4xdvm?postJsllMsg=true

Azure landing zone conceptual architecture

For many organizations, the Azure landing zone conceptual architecture below represents the destination in their cloud adoption journey. It's a mature, scaled-out target architecture intended to help organizations operate successful cloud environments that drive their business while maintaining best practices for security and governance.

This conceptual architecture represents scale and maturity decisions based on a wealth of lessons learned and feedback from customers who have adopted Azure as part of their digital estate.

While your specific implementation might vary, as a result of specific business decisions or existing investments in tools that need to persist in your cloud environment, this conceptual architecture will help set a direction for the overall approach your organization takes to designing and implementing a landing zone.

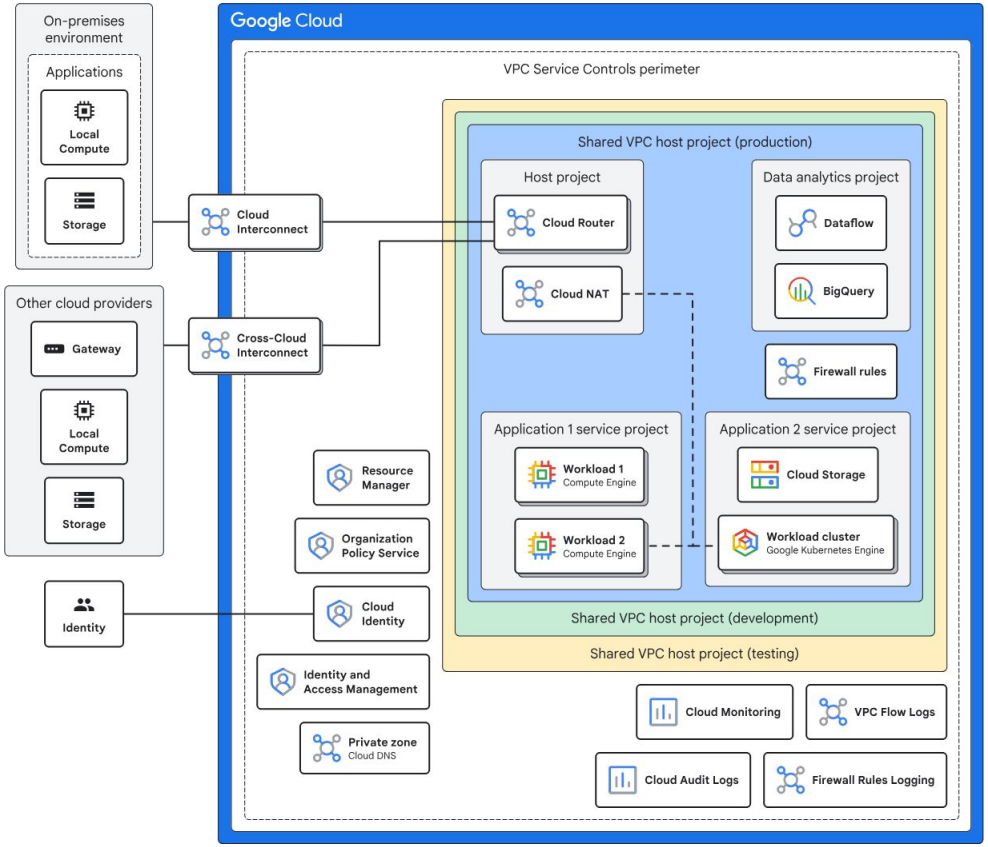
See: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

Landing Zone in GCP

Landing zones help your enterprise deploy, use, and scale Google Cloud services more securely. Landing zones are dynamic and grow as your enterprise adopts more cloud-based workloads over time.

You can deploy a landing zone after you complete the foundational setup for your organization. The foundational setup includes creating an organization resource and creating a billing account, either online or invoiced.

A landing zone spans multiple areas and includes different elements, such as identities, resource management, security, and networking. Many other elements can also be part of a landing zone, as described in Elements of a landing zone.



Sample implementation of a landing zone. It shows an Infrastructure as a Service (IaaS) use case with hybrid cloud and on-premises connectivity in Google Cloud.

See: https://cloud.google.com/architecture/landing-zones

Elements of a Landing Zone in GCP

Elements of a landing zone

A landing zone requires you to design the following core elements on Google Cloud:

- Identity provisioning
- Resource hierarchy
- Network
- Security controls

Sample implementation of a landing zone. It shows an Infrastructure as a Service (IaaS) use case with hybrid cloud and on-premises connectivity in Google Cloud.

See: https://cloud.google.com/architecture/landing-zones

Elements of a Landing Zone in GCP

Landing zone element	Description
Monitoring and logging	Design a monitoring and logging strategy that helps ensure all relevant data is logged and that you have dashboards that visualize the data and alerts that notify you of any actionable exceptions. For more information, see the following: • Set up monitoring, alerting, and logging • Google Cloud operations suite documentation
Backup and disaster recovery	Design a strategy for backups and disaster recovery. For more information, see the following: • Disaster recovery planning guide • Actifio GO for Google Cloud
Compliance	Follow the compliance frameworks that are relevant to your organization. For more information, see the Compliance resource center.

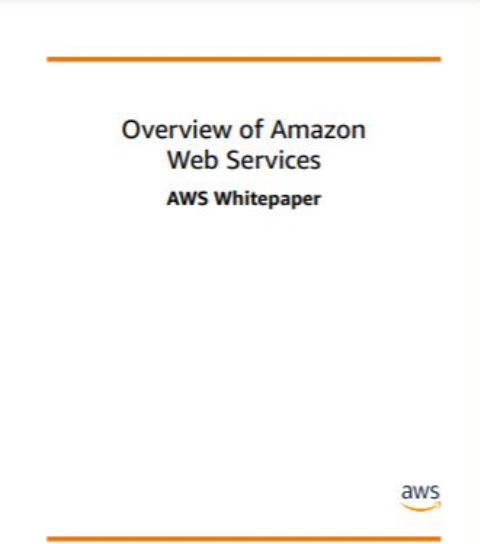
See: https://cloud.google.com/architecture/landing-zones

Elements of a Landing Zone in GCP

Landing zone element	Description
Cost efficiency and control	Design a cost-optimized foundation on Google Cloud and monitor your costs. For more information, see the following: • Overview of cloud billing concepts • Cost optimization on Google Cloud for developers and operators • Cost management
API management	Design a scalable solution for your APIs. For more information, see Apigee API Management.
Cluster management	Design Google Kubernetes Engine (GKE) clusters that follow best practices to build scalable, resilient, and observable services. For more information, see the following: • GKE best practices • Multi-cluster Services • About Anthos Service Mesh

See: https://cloud.google.com/architecture/landing-zones

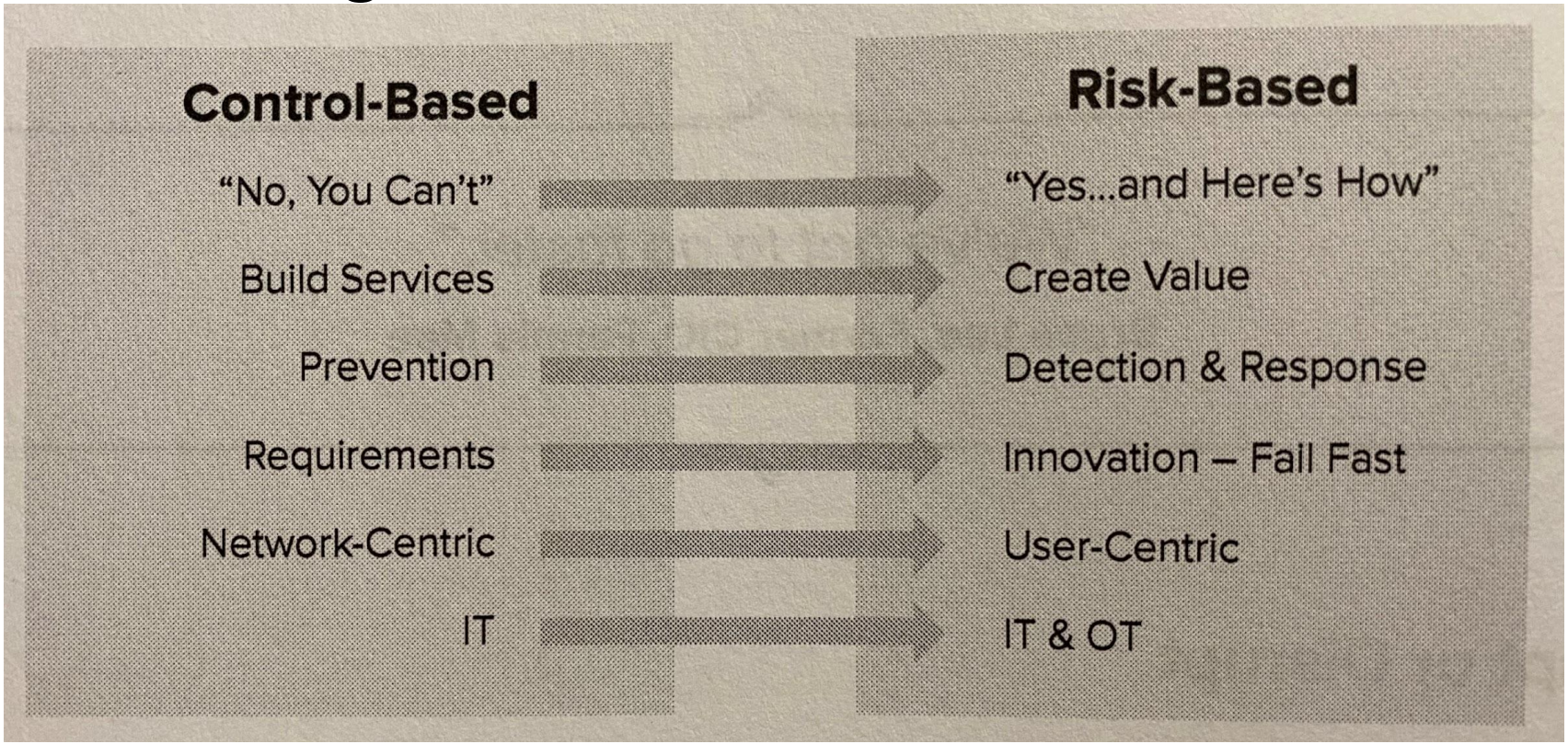
Getting started







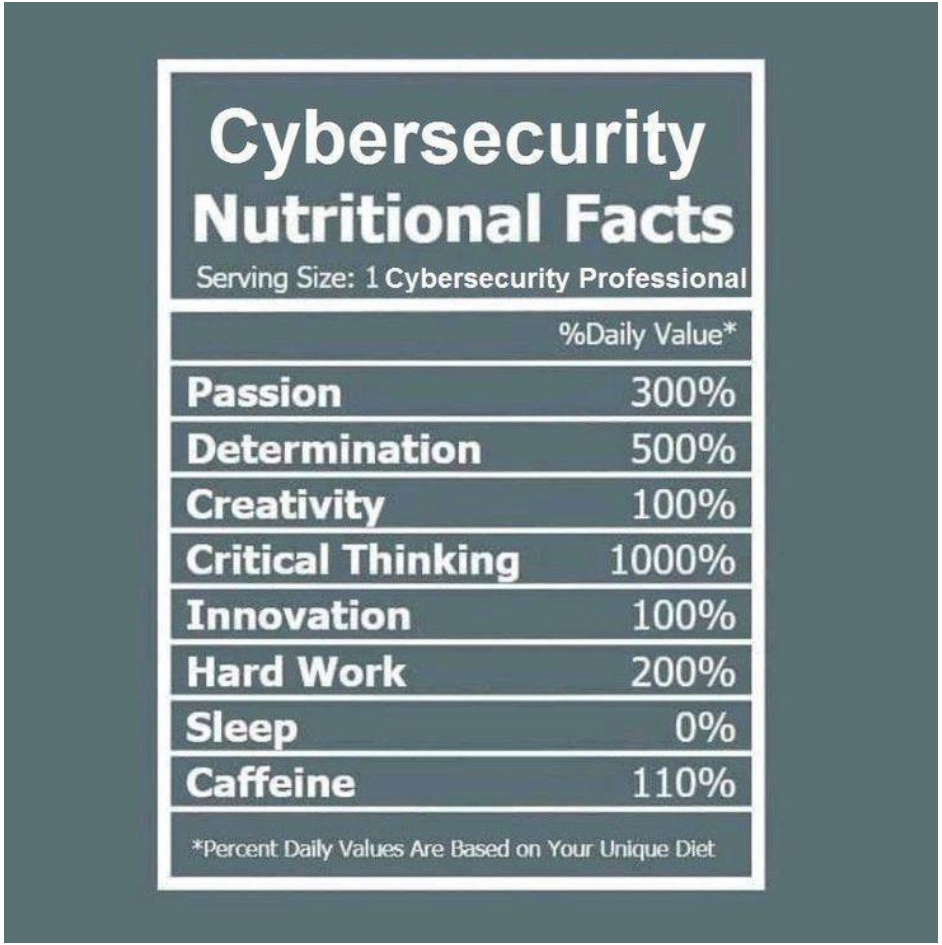
The Change from Control Based to Risk Based



Lucerne University of Applied Sciences and Arts
HOCHSCHULE LUZERN

Source: Stennon, R; Secure Cloud Transformation: The CIO's Journey

Until Next Week!



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